
FR800x Hardware Manual

Bluetooth Low Energy SOC with SIG Mesh

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Date: 2025.5



Revision History

Version	Date	Description
V1.4	2025.5.16	Draft

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1. FR800x Series Introduction

- The FR800x integrates modules such as RF, PMU, QSPI, I2C, UART, GPIO, ADC, PWM, USB OTG, Keyboard scan, and Battery Charger. It supports Bluetooth V5.3 LE and data rates of 2M, 1M, 500K, and 125K. It has a built-in 56KB PSRAM (some chips support 56KB + 2Mb PSRAM) and a 4Mb Flash (the maximum capacity of some models can reach 128Mb). It supports BLE protocols such as GATT, LM, and LC, and also supports the standard SIG Mesh. The display interface supports SPI, QSPI, and MCU8080 - 8bit/16bit.
- A reasonable PCB layout, clear traces, and good grounding are the basic requirements for a good design. The FR800x has fewer peripheral components, so a double-sided PCB can be used for design to save costs. In the double-sided PCB design, the top layer is used for placing components and routing signal lines, while the bottom layer is used for routing power lines and ensuring a complete ground plane as much as possible.

FR800x series:

- FR8003A: built-in 4Mb Flash and packaged in QFN20_L3×W3×T0.75_P0.4mm.
- FR8003D: built-in 8Mb Flash and packaged in QFN20_L3×W3×T0.75_P0.4mm.
- FR8008AP: built-in 4Mb Flash and packaged in QFN40_L5×W5×T0.9_P0.4.
- FR8008FP: built-in 32Mb Flash and packaged in QFN40_L5×W5×T0.9_P0.4.
- FR8008GP: built-in 64Mb Flash and packaged in QFN40_L5×W5×T0.9_P0.4.
- FR8008HP: built-in 128Mb Flash and packaged in QFN40_L5×W5×T0.9_P0.4.
- FR8008A: built-in 4Mb Flash and packaged in QFN48_L6×W6×T0.75_P0.4

- Application Fields:

- Smart Bracelet
- Smart Home
- Knob Screen of Home Appliances
- USB Dongle
- Remote Controller
- Multi-connection Private Network Deployment

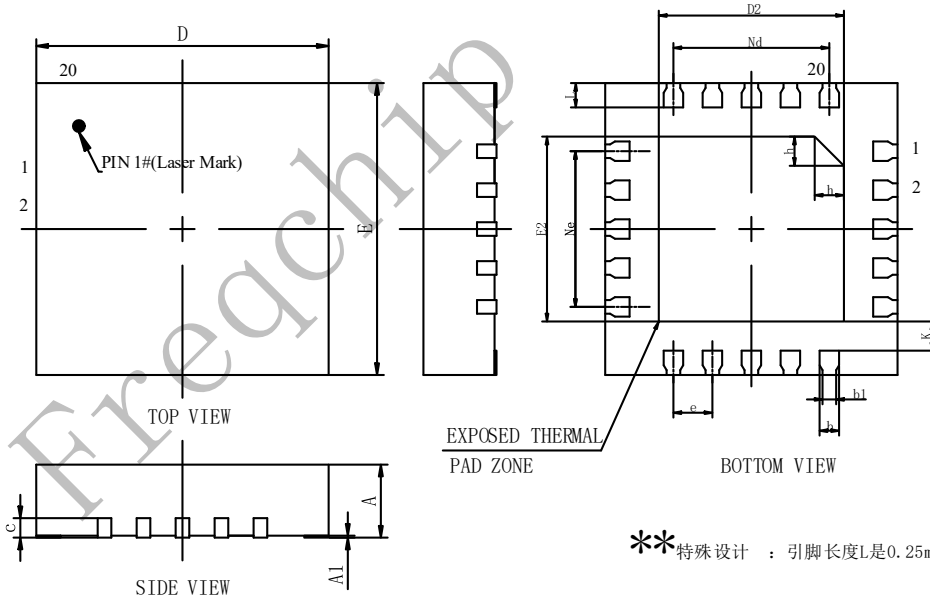
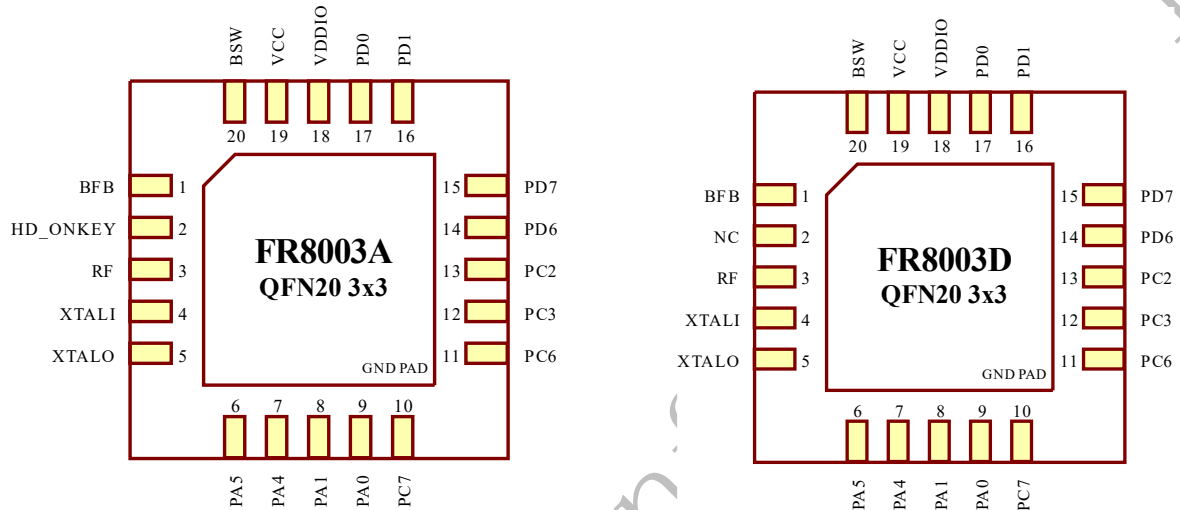
- Mesh Network Deployment
- Mouse
- Keyboard
- Smart Door Lock
- Data Transparent Transmission Module

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2. Hardware Information

2.1. FR8003x Pinout and Package

- FR8003A/D: QFN20 3x3

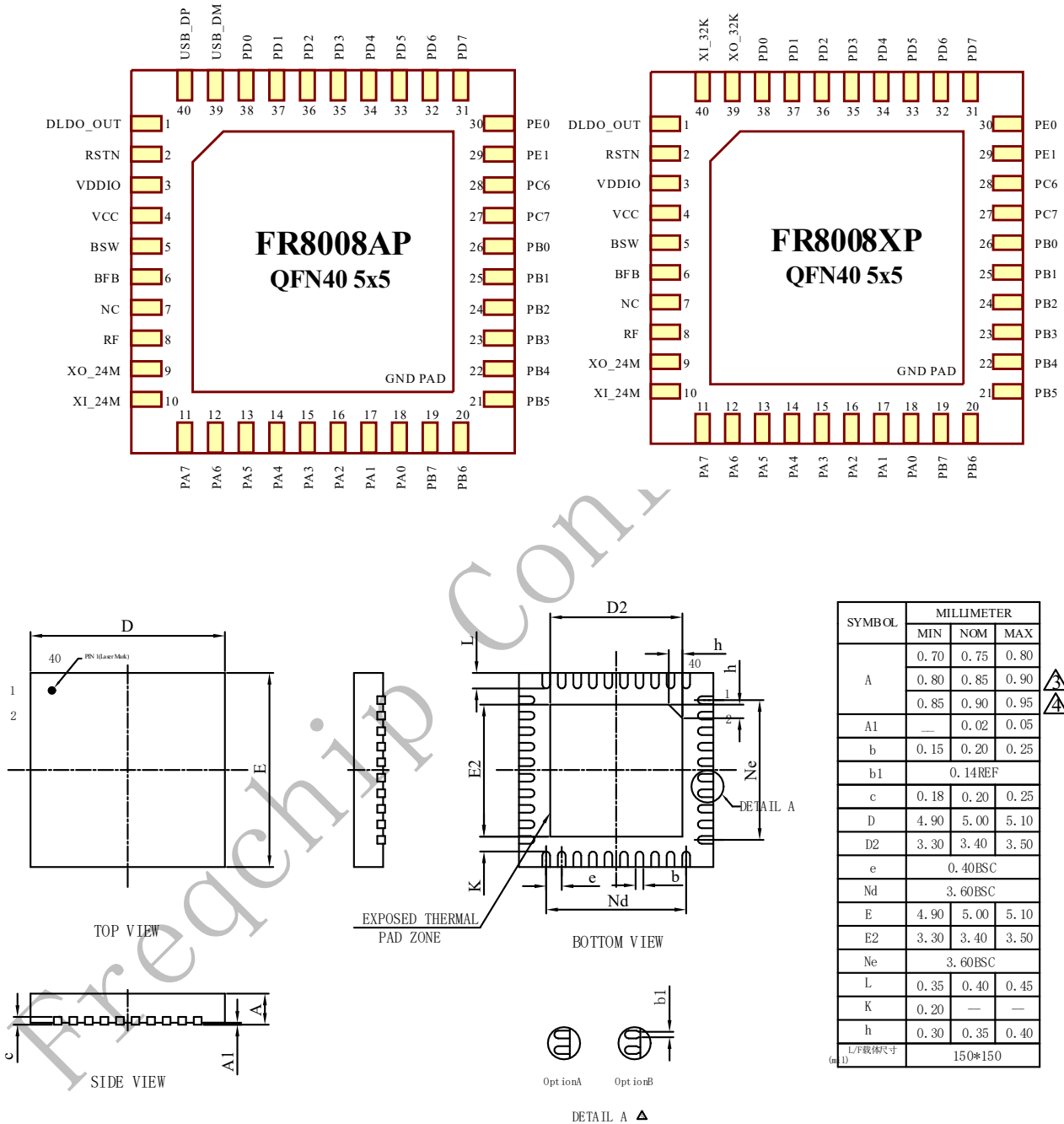


SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
	0.85	0.90	0.95
A1	0	0.02	0.05
b	0.15	0.20	0.25
b1	0.14REF		
c	0.203REF		
D	2.90	3.00	3.10
D2	1.80	1.90	2.00
e	0.40BSC		
Ne	1.60BSC		
Nd	1.60BSC		
E	2.90	3.00	3.10
E2	1.80	1.90	2.00
L	0.20	0.25	0.30
h	0.25	0.30	0.35
K	0.30REF		

**特殊设计：引脚长度L是0.25mm；

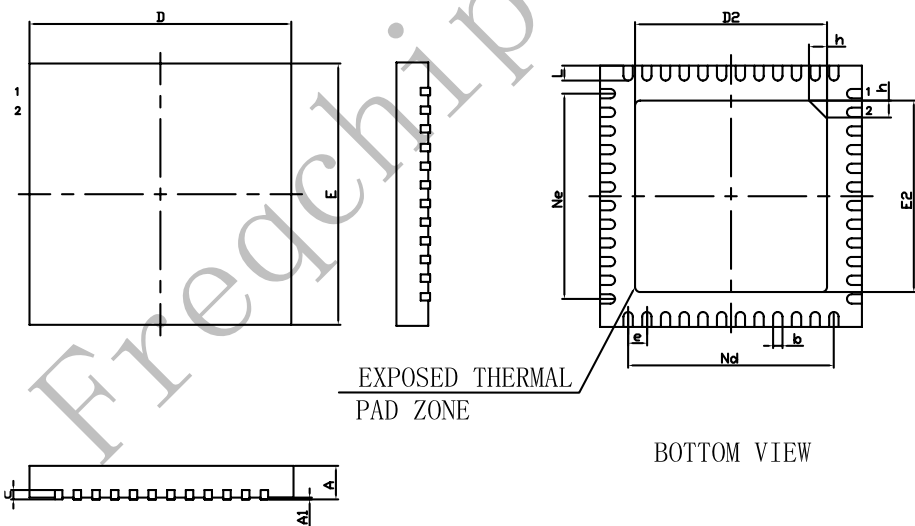
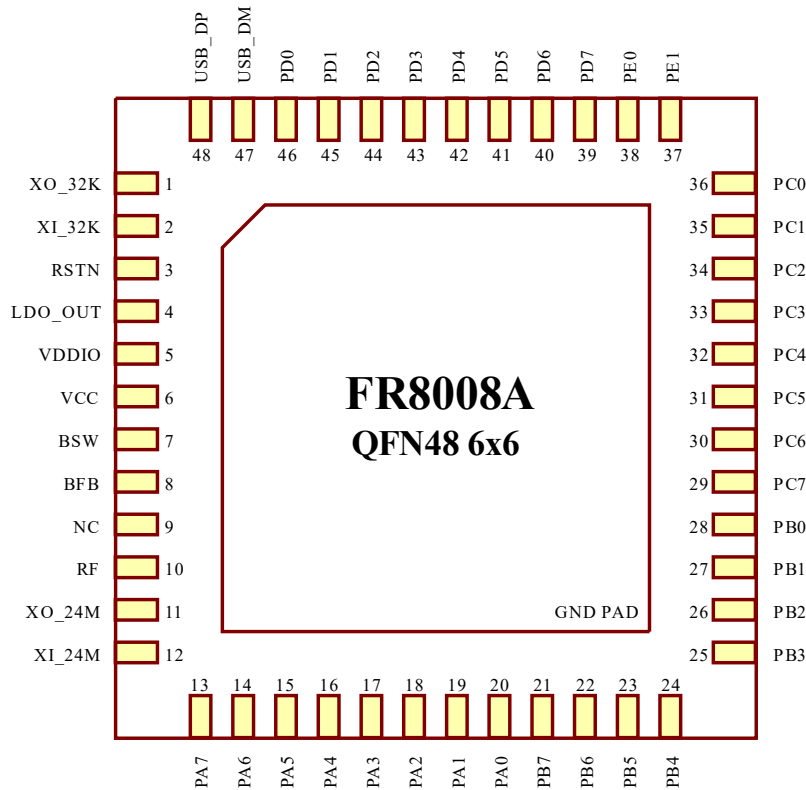
2.2. FR8008XP Pinout and Package

- FR8008AP/XP: QFN40 5x5



2.3. FR8008A Pinout and Package

● FR8008A: QFN48 6x6



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	—	0.02	0.05
b	0.15	0.20	0.25
c	0.18	0.20	0.23
D	5.90	6.00	6.10
D2	4.10	4.20	4.30
e	0.40BSC		
Ne	4.40BSC		
Nd	4.40BSC		
E	5.90	6.00	6.10
E2	4.10	4.20	4.30
L	0.35	0.40	0.45
h	0.30	0.35	0.40
L/P载体尺寸 (mm)	177*177		

2.4. FR800x Pins Description

Notation	Description
AI	Analog input
AO	Analog output
I/O	Bidirectional(digital)
PWR	Power
GND	Ground

Pins Number					Name	Type	Description
FR8008A	FR8008AP	FR8008XP	FR8003A	FR8003D			
	1	1			DLDO_OUT	PWR	Linear regulator output with on/off switch control
3	2	2			RSTN	AI	Reset pin, active at low level
5	3	3	18	18	VDDIO	PWR	I/O power supply, capable of providing external power output (1.8-3.3V/80mA)
6	4	4	19	19	VCC	PWR	Power supply (1.8-3.3V)
7	5	5	20	20	BSW	AO	BUCK switch output
8	6	6	1	1	BFB	AI	BUCK feedback input
9	7	7		2	NC	-	Not connected
10	8	8	3	3	RF	AIO	RF input and output
12	10	10	4	4	XTALI_24M	AI	24MHz Crystal oscillator input
11	9	9	5	5	XTALO_24M	AO	24MHz Crystal oscillator output
13	11	11			PA7	I/O	PA7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
14	12	12			PA6	I/O	PA6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
15	13	13	6	6	PA5	I/O	PA5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM

Pins Number					Name	Type	Description
FR8008A	FR8008AP	FR8008XP	FR8003A	FR8003D			
16	14	14	7	7	PA4	I/O	PA4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / I2S.BCLK
17	15	15			PA3	I/O	PA3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / I2S.MISO
18	16	16			PA2	I/O	PA2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / I2S.MOSI
19	17	17	8	8	PA1	I/O	PA1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / BLE.RX / SIROUT / I2S.FRM
20	18	18	9	9	PA0	I/O	PA0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / BLE.TX / SIRIN / I2S.BCLK
21	19	19			PB7	I/O	PB7 / I2C1.DAT / SPIM0.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / CLK.OUT / RS485.EN / I2S.MISO
22	20	20			PB6	I/O	PB6 / I2C1.CLK / SPIM0.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / ANT.RX / SIROUT / I2S.MOSI
23	21	21			PB5	I/O	PB5 / I2C0.DAT / SPIM0.IO3 / SPIS.CSN / UTXD0 / USBDM / PWM5 / PDM.DAT / ANT.TX / SIRIN / I2S.FRM
24	22	22			PB4	I/O	PB4 / I2C0.CLK / SPIM0.IO2 / SPIS.CLK / URXD0 / USBDP / PWM4 / PDM.CLK / CLK.OUT / LCD.DCX / I2S.BCLK
25	23	23			PB3	I/O	PB3 / I2C1.DAT / SPIM0.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / WLAN.RX / LCD.CSX / BURN.SPIMISO
26	24	24			PB2	I/O	PB2 / I2C1.CLK / SPIM0.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / WLAN.TX / RS485.EN / BURN.SPIMOSI

Pins Number					Name	Type	Description
FR8008A	FR8008AP	FR8008XP	FR8003A	FR8003D			
27	25	25			PB1	I/O	PB1 / I2C0.DAT / SPIM0.CSN / SPIS.CSN / UTXD0 / USBDM / PWM1 / PDM.DAT / BLE.RX / SIROUT / BURN.SPICSN
28	26	26			PB0	I/O	PB0 / I2C0.CLK / SPIM0.CLK / SPIS.CLK / URXD0 / USBDP / PWM0 / PDM.CLK / BLE.TX / SIRIN / BURN.SPICLK
29	27	27	10	10	PC7	I/O	PC7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SWDIO / RS485.EN / I2S.MISO
30	28	28	11	11	PC6	I/O	PC6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SWTCK / SIROUT / I2S.MOSI
37	29	29			PE1	I/O	PE1 / I2C0.DAT / SPIM1.IO1 / SPIS.CSN / UTXD0 / UTXD1 / PWM1 / PDM.DAT / BLE.RX / SIROUT / USBDM
38	30	30			PE0	I/O	PE0 / I2C0.CLK / SPIM1.IO0 / SPIS.CLK / URXD0 / URXD1 / PWM0 / PDM.CLK / BLE.TX / SIRIN / USBDP
39	31	31	15	15	PD7	I/O	PD7 / I2C1.DAT / SPIM1.CSN / SPIS.MISO / UCTS0 / UTXD1 / PWM7 / PDM.DAT / SARADC0 / RS485.EN / I2S.MISO
40	32	32	14	14	PD6	I/O	PD6 / I2C1.CLK / SPIM1.CLK / SPIS.MOSI / URTS0 / URXD1 / PWM6 / PDM.CLK / SARADC1 / SIROUT / I2S.MOSI
41	33	33			PD5	I/O	PD5 / I2C0.DAT / SPIM1.IO3 / SPIS.CSN / UTXD0 / PWM5 / PDM.DAT / SARADC2 / SIRIN / I2S.FRM
42	34	34			PD4	I/O	PD4 / I2C0.CLK / SPIM1.IO2 / SPIS.CLK / URXD0 / PWM4 / PDM.CLK / SARADC3 / I2S.BCLK
43	35	35			PD3	I/O	PD3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SARADC4 / I2S.MISO

Pins Number					Name	Type	Description
FR8008A	FR8008AP	FR8008XP	FR8003A	FR8003D			
44	36	36			PD2	I/O	PD2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SARADC5 / RS485.EN / I2S.MOSI
45	37	37	17	17	PD1	I/O	PD1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / PWM1 / PDM.DAT / SARADC6 / SIROUT / I2S.FRM
46	38	38	16	16	PD0	I/O	PD0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / PWM0 / PDM.CLK / SARADC7 / SIRIN / I2S.BCLK
47	39		-	-	USB_DM	I/O	USB Digital Minus
48	40		-	-	USB_DP	I/O	USB Digital Positive
1		39			XTALO_32K	AO	32KHz Crystal oscillator output
2		40			XTALI_32K	AI	32KHz Crystal oscillator input
			2		ONKEY	AI	Wake up input pin while in power off mode, the current is less than 1uA
31					PC5	I/O	PC5 / I2C0.DAT / SPIM1.IO3 / SPIS.CSN / UTXD0 / LCD.RDX / PWM5 / PDM.DAT / SWV / SIRIN / I2S.FRM / PSRAM.IO0 / LCD.D13
32					PC4	I/O	PC4 / I2C0.CLK / SPIM1.IO2 / SPIS.CLK / URXD0 / LCD.WRX / PWM4 / PDM.CLK / ANT.RX / LCD.TE / I2S.BCLK / PSRAM.IO2 / LCD.D12
33			12	12	PC3	I/O	PC3 / I2C1.DAT / SPIM1.IO1 / SPIS.MISO / UCTS0 / UTXD1 / PWM3 / PDM.DAT / SWV / I2S.MISO
34			13	13	PC2	I/O	PC2 / I2C1.CLK / SPIM1.IO0 / SPIS.MOSI / URTS0 / URXD1 / PWM2 / PDM.CLK / SWV / RS485.EN / I2S.MOSI
35					PC1	I/O	PC1 / I2C0.DAT / SPIM1.CSN / SPIS.CSN / UTXD0 / LCD.DCX / PWM1 / PDM.DAT / SWV / SIROUT / I2S.FRM / PSRAM.CLK / LCD.D9
36					PC0	I/O	PC0 / I2C0.CLK / SPIM1.CLK / SPIS.CLK / URXD0 / LCD.CSX / PWM0 / PDM.CLK / SWV / SIRIN / I2S.BCLK / PSRAM.IO3 / LCD.D8

3. Application Circuit

3.1. FR8003A Circuit

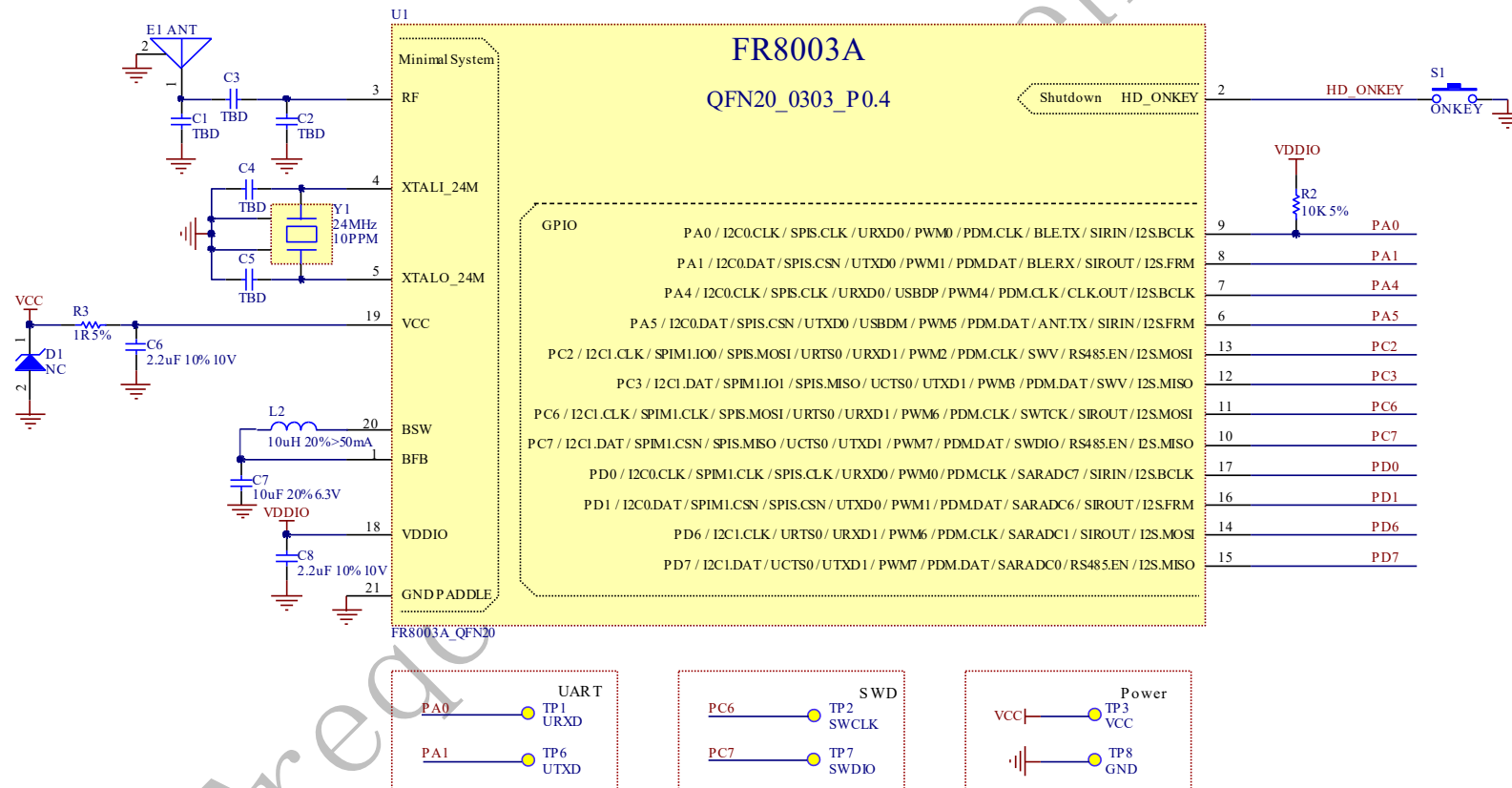


Figure 3-1 FR8003A Circuit

3.2. FR8003D Circuit

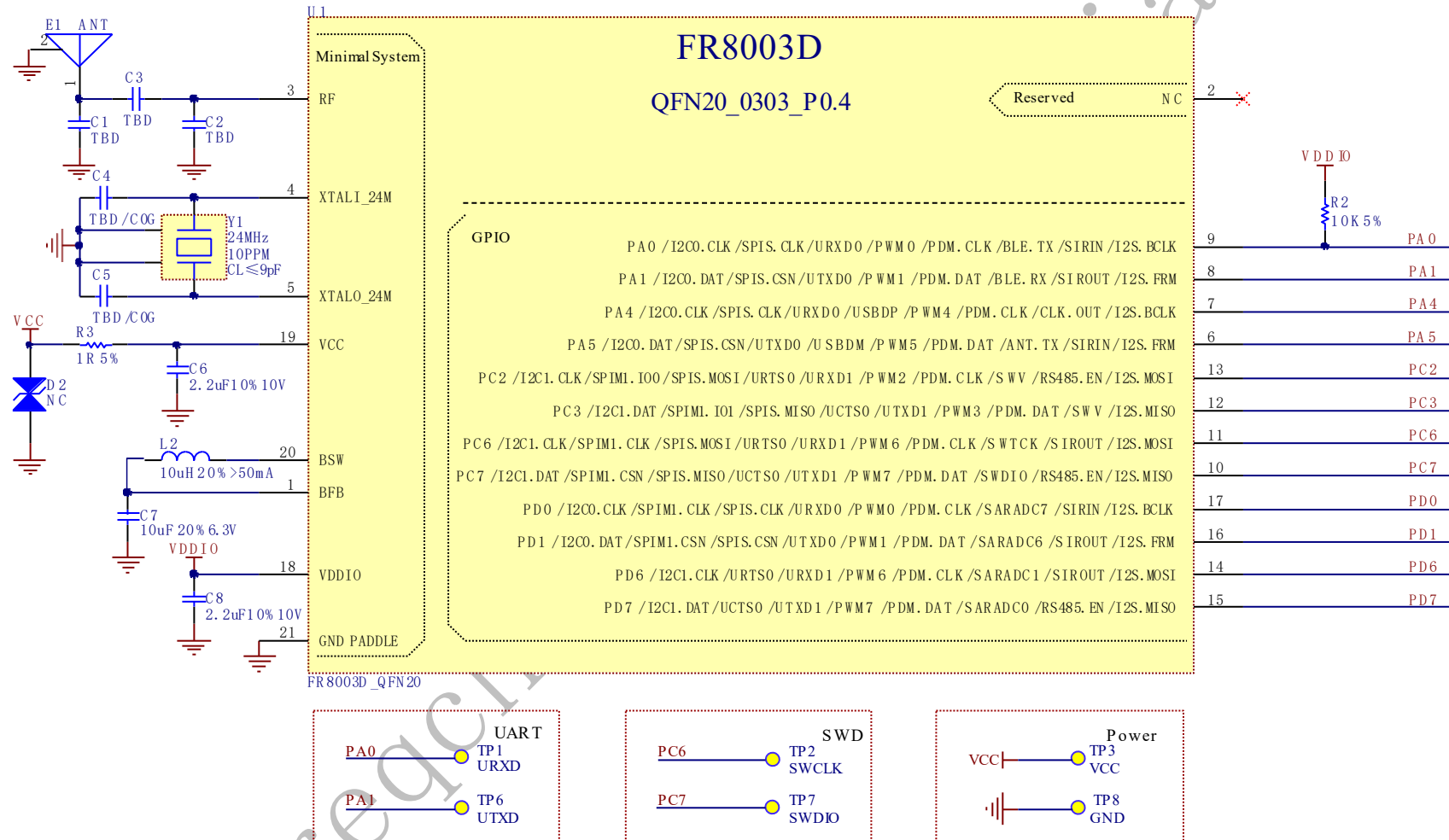


Figure 3-2 FR8003D Circuit

3.3. FR8008AP Circuit

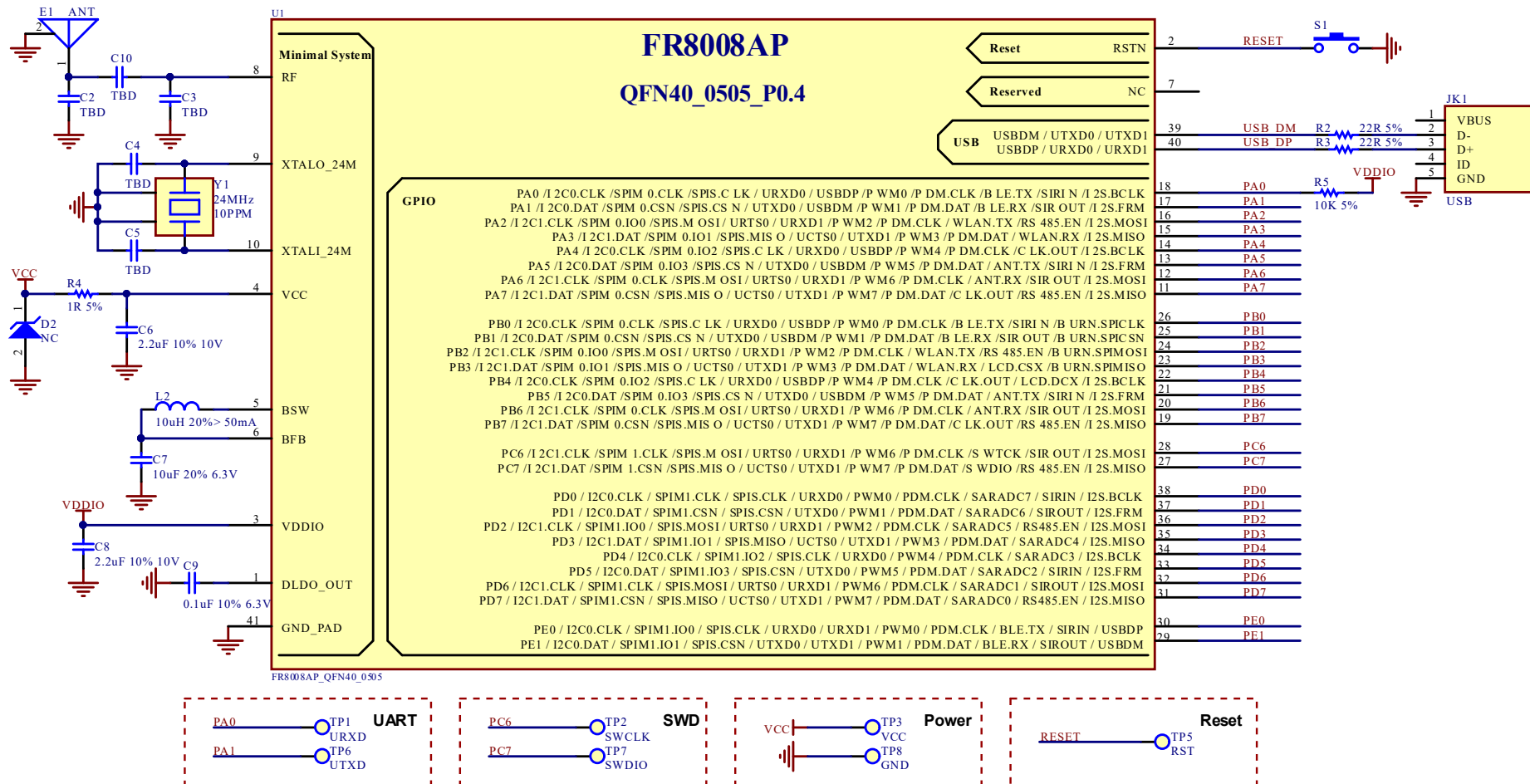


Figure 3-3 FR8008AP Circuit

3.4. FR8008XP Circuit

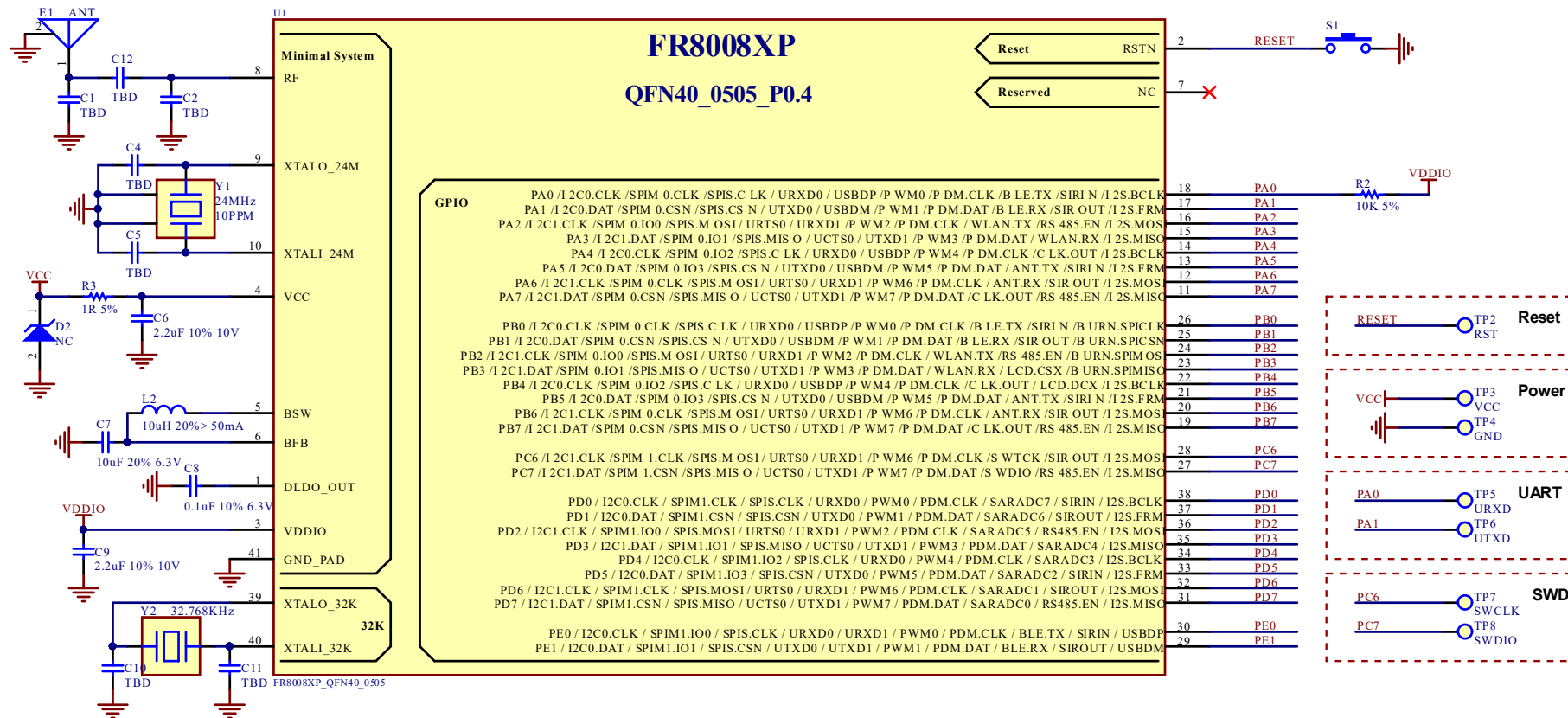


Figure 3-4 FR8008XP Circuit

3.5. FR8008A Circuit

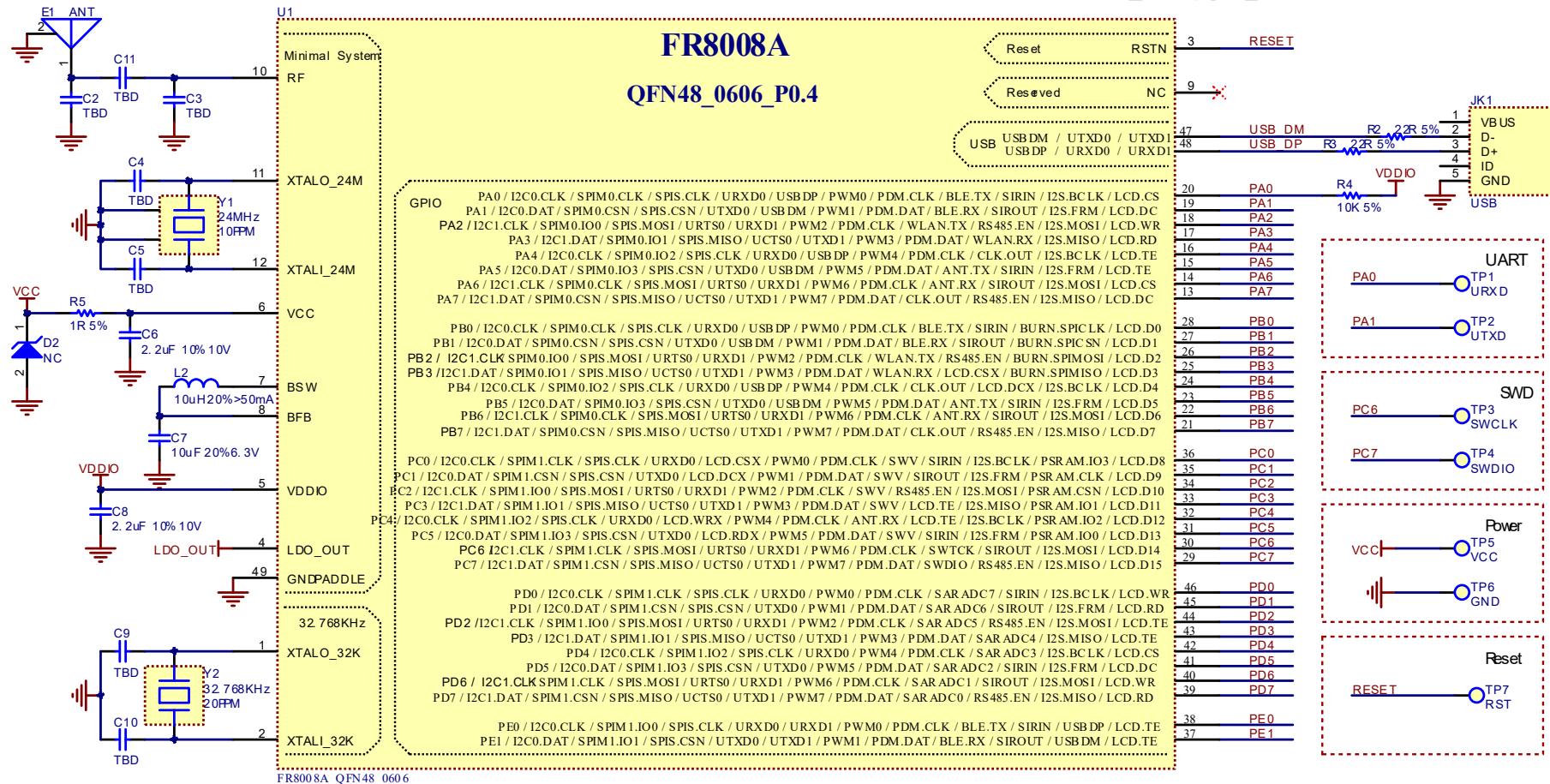


Figure 3-5 FR8008A Circuit

4. Application Design

4.1. Schematic Design

4.1.1. Component Selection

- 1) For the inductor in the antenna matching circuit, select a high-frequency inductor with a high Q factor and low internal resistance. For the capacitor, choose a Class I capacitor (NP0/C0G) with good stability.
- 2) It is recommended to use a crystal oscillator with a frequency tolerance of ± 10 PPM and $CL \leq 12$ pF. Reserve positions for crystal oscillator load capacitors and determine the capacitance values through frequency offset testing.
- 3) For L2, select a wire-wound inductor or a power multilayer inductor with a rated current greater than 50 mA, internal resistance less than 1 ohm, and self-resonant frequency greater than 10 MHz.
- 4) The capacitor voltage rating for the VBAT pin should be no less than 10 V, and for the VCHG pin, no less than 16 V.

4.1.2. Pin Descriptions

- 1) The FR8008XP supports functions such as UART, I2C, SPI, PWM, ADC, I2S, and PDM.
- 2) PA0 and PA1 are UART ports by default, while PC6 and PC7 are J-Link ports by default. Test points must be reserved on the PCB for PA0 and PA1. If space allows on the PCB, test points should also be reserved for PC6 and PC7.
- 3) Port D supports SAR ADC, with PD6 and PD7 supporting ADKEY.
- 4) All GPIO ports support output, input, and floating states, with the default being the floating state.
- 5) The maximum drive current when an I/O port outputs a high level is 8 mA.
- 6) The RSTN pin is low-level reset.
- 7) The maximum output current of the VDDIO pin is 100 mA @ 2.9 V

4.2. PCB Layout Considerations

Circuit Design and Layout:

- (1) The upper left part is the minimum system of FR8008XP. In non-low-power applications, the internal LDO of the chip can be used for voltage step-down, eliminating the need for inductor L2 and reducing the value of C7 (not less than 2.2uF).
- (2) Keep the antenna area clear, away from metal and batteries, and away from high-frequency noise sources. Reserve a π -type matching circuit for antenna impedance debugging.
- (3) RF traces should be designed as 50 Ω microstrip lines, with the shortest possible distance to the antenna. Use arcs or 135° corners instead of right angles. Connect the antenna reference GND to the chip bottom GND via copper pour, avoiding segmented by other traces. Densely place GND vias around RF traces for shielding.
- (4) Place L2 close to the BSW pin of the chip, with copper pour on the adjacent layer. Position C7 close to L2, ensuring the BFB trace passes through C7 with short and wide routing. Place at least two GND vias at the negative terminal of C7, connecting to the chip bottom GND via copper pour without segmented by other traces.
- (5) Do not place inductor L2 and crystal Y1 too close; keep a sufficient distance to avoid affecting RF performance. Place dense GND vias around the crystal for shielding.
- (6) Place all voltage-stabilizing and filtering capacitors close to the relevant chip pins.
- (7) If the product needs to pass certifications such as IEC with ESD requirements >2KV, reserve ESD protection devices at relevant pins and interfaces.

4.3. GPIO Considerations

All pins can be configured in two states: input and output. The high-level output of GPIO pins is the voltage of the VDDIO pin, and the voltage value of the VDDIO pin can be configured via API. When GPIO is in input mode, the thresholds are as follows: the high level needs to be $> 0.7 * VDDIO$ pin voltage, and the low level needs to be $< 0.3 * VDDIO$ pin voltage.

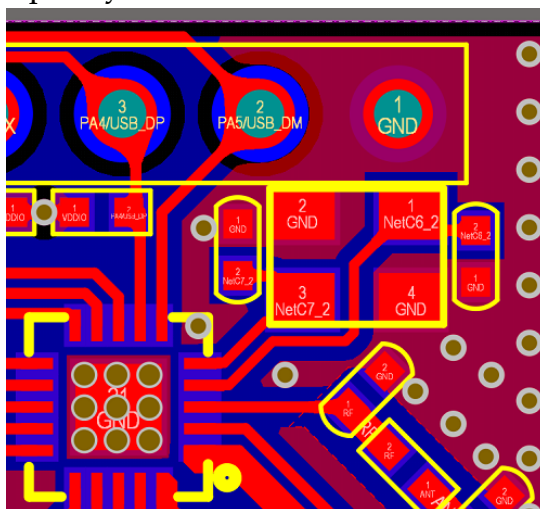
Note: During system initialization and operation, the built-in rom code configures pins PA0 and PA1 as control pins for UART0. PA1 outputs internal UART print information, while PA0 receives external print information. During design, it is advisable to connect buttons or high-impedance components to avoid affecting software programming.

4.4. RF Circuit

- RF traces must be 50Ω impedance-controlled lines. Keep traces as short and wide as possible, avoid vias, and preferably route on the same layer as the chip. Do not use right angles; use Arc-shaped traces or 135° angles instead. Maintain consistent trace width, avoid branching, and shield with surrounding ground vias. Ensure a continuous ground plane beneath RF traces, connected to the chip's bottom GND.
- Reserve a π -type matching circuit close to the chip's RF pin for antenna impedance matching. Adjust antenna impedance toward 50Ω. Component values are determined via actual measurements based on the antenna and PCB layout.
- Use an inverted-F antenna whenever possible.
- Place the antenna at the board edge, never enclosed by GND. Maintain clearance zones on both sides, away from metal, batteries, and high-frequency noise sources

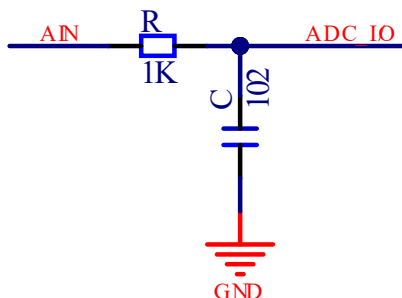
the FR800x. A surface-mount crystal oscillator with a grounded casing is recommended for more stable performance.

- In practical applications, even with the same components, frequency deviation may occur due to differences in layout and PCB materials. Therefore, before mass production of each design, several prototype units must be tested to measure frequency deviation. Adjust the crystal oscillator load capacitance or corresponding register values to control the frequency deviation within $\pm 30\text{KHz}$



4.6. ADC I/O Usage Instructions

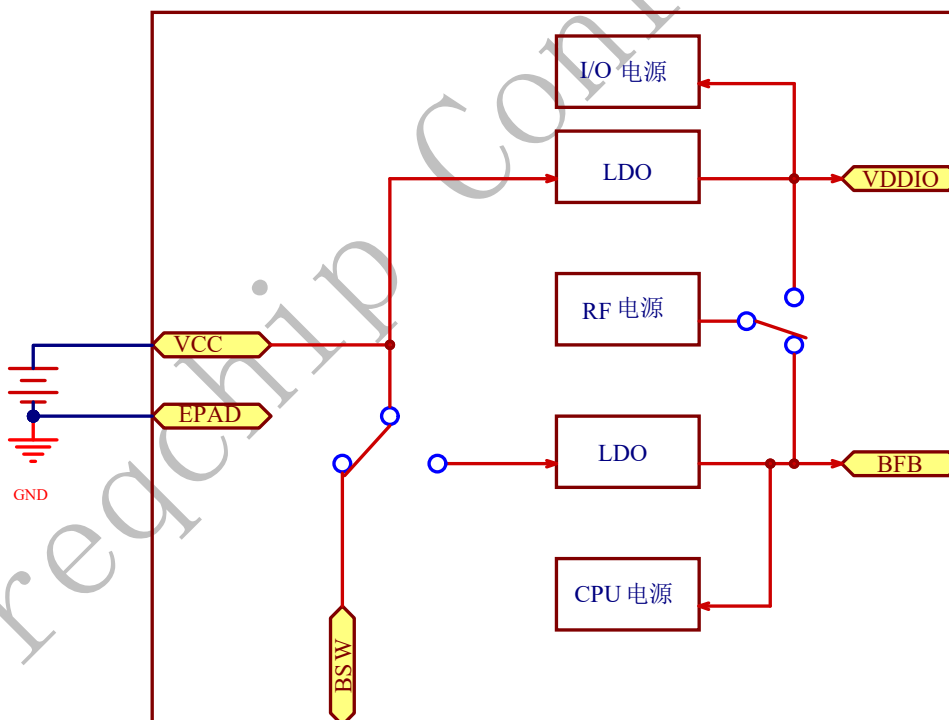
- ADC uses I/O pins: PD0~PD7
- The ADC signal acquisition circuit is shown in the following figure:



- It is recommended to use a voltage divider circuit with a ratio of 1/2 or 1/3. The input voltage of the I/O should be lower than the VDDIO voltage. The reference voltage for the ADC is 1.2V or the VDDIO voltage.

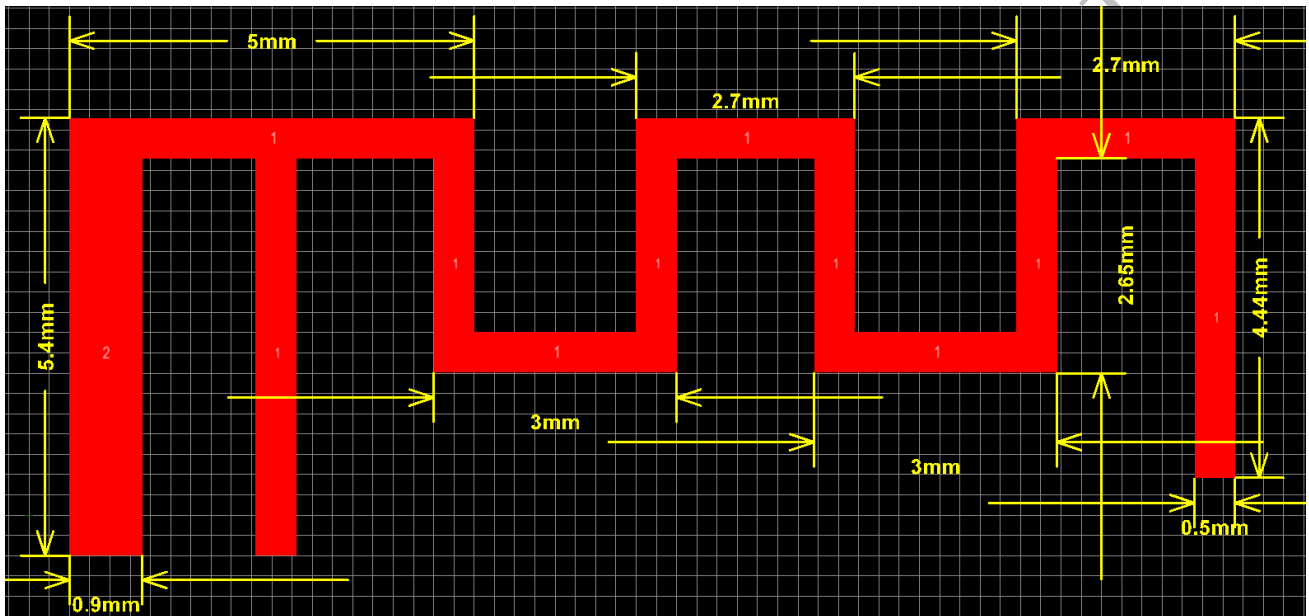
4.7. System Design

- Before mass production of each design, several prototype units shall be sampled for frequency offset debugging and antenna impedance matching based on the actually used crystal oscillator and antenna. If frequency offset and antenna impedance matching debugging are not performed, the RF performance will be affected, manifested as long search and connection time, unstable data transmission and reception, disconnection, etc.
- In systems with high-power devices such as motors, during PCB design, the GND loop of the Bluetooth part must be separated from that of high-power devices, and then connected to the power ground at a single point.
- The internal power supply system of the chip is shown in the following figure, with only VBAT as the main power input



5. PCB Reference Antenna

- The PCB onboard antenna can meet general application requirements but has slightly inferior performance and is prone to interference. Its advantages include low cost and convenient assembly. For applications with small product size and high-performance requirements, surface-mount ceramic antennas or external antennas are recommended.
- Inverted-F Antenna.



Contact Information

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Feedback:

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